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BALLARI INSTITUTE OF TECHNOLOGY & MANAGEMENT

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"JnanaGangotri" Campus, No.873/2, Ballari-Hospet Road, Allipur, Ballar1-583 104 (Karnataka) (India)
Ph: 08392 – 237100 / 237190, Fax: 08392 – 237197

DEPARTMENT OF CSE(DATA SCIENCE)

A Mini-Project Report On

"STATE-WISE RAINFALL ANALYSIS DASHBOARD "

A report submitted in partial fulfillment of the requirements for the

DATA VISUALIZATION WITH POWER BI

Submitted By

SHIRISHA TS **USN: 3BR23CD088**

Under the Guidance of
Mr.Sandeep Sindhanoor
Asst. Professor

Dept of CSE (DATA SCIENCE),
BITM, Ballari

Visvesvaraya Technological University

Belagavi, Karnataka 2025-2026

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CERTIFICATE

This is to certify that the OPEN-ENDED PROJECT of DATA VISUALIZATION WITH POWER BI **"STATE-WISE RAINFALL ANALYSIS DASHBOARD"** has been successfully presented by **SHIRISHA T S** bearing USN **3BR23CD088** student of semester B.E for the partial fulfillment of the requirements for the award of **Bachelor Degree in CSE(DS)** of the BALLARI INSTITUTE OF TECHNOLOGY& MANAGEMENT, BALLARI during the academic year 2025-2026.

Signature of guide	Signature of HOD
Mr. Sundeep Sindhanoor	**Dr. Aradhana D**

ACKNOWLEDGEMENT

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Name	USN
SHIRISHA T S	3BR23CD088

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ABSTRACT

The **“Rainfall Analysis Dashboard – State-wise”** project is designed to provide a comprehensive, interactive, and user-friendly visualization of rainfall patterns across different states in India using Power BI. It transforms raw rainfall data into meaningful insights through visuals such as maps, bar charts, line graphs, and KPI indicators, enabling users to understand both spatial distribution and temporal trends of rainfall. The dashboard allows dynamic filtering by state, district, and time period, making it easy to compare regions, identify high rainfall zones, and analyze seasonal variations. It also includes key metrics like total, average, maximum, and minimum rainfall to give quick and clear insights. By supporting trend analysis and pattern recognition, the system helps in understanding monsoon behavior and climate variability. Additionally, it can assist in planning and decision-making related to agriculture, water resource management, and disaster preparedness by highlighting potential risks such as floods or droughts. Overall, the dashboard converts complex environmental data into an accessible and visually engaging format, making it useful for researchers, policymakers, and students to make informed, data-driven decisions.

CHAPTER 1

INTRODUCTION

1.1 About the Project

The **"High Rainfall Dashboard – State-wise"** project focuses on developing a dynamic and interactive Power BI dashboard to monitor and analyze rainfall patterns across different states in India. As shown in the uploaded dashboard, it includes multiple visual components such as a **map interface, bar charts, pie charts, and monthly trend graphs**, which together provide a clear and comprehensive understanding of rainfall distribution across regions.

Using the capabilities of Power BI, this project transforms raw rainfall data into meaningful and visually appealing insights that are easy to interpret. The dashboard enables users to analyze rainfall at different levels, including **state, station name, season, and month**, through interactive filters. It helps identify high rainfall areas, compare rainfall distribution across states, and understand seasonal and monthly variations.

The importance of rainfall data analysis is significant in areas such as agriculture, climate monitoring, and disaster management. This project aggregates and visualizes key metrics like **total rainfall, average rainfall, and monthly rainfall trends**, allowing users to track changes over time. As seen in the dashboard, visualizations such as **"Sum of Rainfall by Month," "Rainfall by State," and map-based distribution** provide deeper insights into rainfall patterns and regional differences.

The dashboard supports real-time interaction, enabling users to filter and explore data dynamically based on various parameters such as state, district, station, and season. This enhances the user's ability to perform detailed analysis and gain a better understanding of rainfall behavior across different regions.

CHAPTER 1

1.2 Language Used

The core tools and technologies used in this project are:

- **Power BI**: Used for data modeling, dashboard creation, and visualization

- **DAX (Data Analysis Expressions)**: Used to create measures such as total rainfall and average rainfall

- **Power Query (M Language)**: Used for data cleaning and transformation

- **Microsoft Excel / CSV**: Used as the data source for importing rainfall datasets

These tools together provide a complete solution, enabling efficient data processing, analysis, and visualization in an interactive and user-friendly manner.

CHAPTER 2

SCOPE OF THE PROJECT

2.1 Problem Statement

Large volumes of rainfall data are generated across different states and regions in India through weather monitoring systems. However, this data is often available in raw or tabular formats, making it difficult for users—especially non-technical individuals—to interpret patterns, trends, and regional variations effectively. The absence of interactive and visual tools limits the ability of researchers, policymakers, and students to analyze rainfall distribution, identify high rainfall zones, and understand seasonal changes.

This project addresses these challenges by developing an interactive Power BI dashboard that visually represents rainfall data across states. It enables users to easily monitor, compare, and analyze rainfall trends through maps, charts, and dynamic filters.

2.2 Objectives

The main objectives of the project are as follows:

- To collect and integrate rainfall data at the state and station level.
- To clean and transform raw data into a structured format suitable for analysis.
- To design interactive dashboards that present rainfall distribution using maps, bar charts, and line graphs.
- To identify high rainfall regions and analyze monthly and seasonal rainfall trends.
- To provide meaningful insights through KPIs such as total rainfall, average rainfall, and rainfall distribution.

CHAPTER 3

SYSTEM REQUIREMENTS

3.1 Functional Requirements

- Load and refresh rainfall datasets from Excel/CSV files.
- Filter data by **state, district, station name, season, and month**.
- Display interactive visualizations such as **maps, bar charts, pie charts, and line graphs**.
- Show rainfall distribution across different regions using a **map interface**.
- Provide KPI indicators like **total rainfall, average rainfall, and monthly trends**.
- Enable users to compare rainfall across states and analyze seasonal variations.

3.2 Non-Functional Requirements

- Fast response time when applying filters and interacting with visuals.
- User-friendly interface with clear layout and easy navigation.
- High performance even with large rainfall datasets.
- Scalable system that can handle additional data such as temperature or wind speed in the future.

CHAPTER 3

3.3 Hardware Requirements

- **Processor**: Intel i5 or higher
- **RAM**: 8 GB or more
- **Storage**: Minimum 500 MB for project files
- **Display**: Full HD recommended for better visualization

3.4 Software Requirements

- **Power BI Desktop** – For dashboard creation and visualization
- **Microsoft Excel / CSV** – For data source and preprocessing
- **Windows 10 or later**

CHAPTER 4

METHODOLOGY

The methodology of the **“High Rainfall Dashboard”** project focuses on transforming raw rainfall data into meaningful visual insights using Power BI. Instead of complex database modeling or ER diagrams, this project follows a simple and practical approach consisting of data collection, data preparation, and visualization.

4.1 Data Collection

The dataset used in this project contains rainfall-related information such as:

- State
- District
- Station Name
- Month and Year
- Season
- Rainfall values
- Temperature (optional)

This data is collected and stored in Excel/CSV format, which is then imported into Power BI for analysis.

4.2 Data Preparation (Power Query)

After loading the dataset into Power BI, data cleaning and transformation are performed using Power Query:

- Removed missing or null values
- Standardized column names
- Converted date-related fields (Month, Year)
- Organized data for better analysis

This step ensures that the dataset is accurate and ready for visualization.

CHAPTER 4

4.3 Data Visualization

The cleaned data is then used to create interactive visualizations in Power BI. As shown in the dashboard:

Visuals Used:

- **Map Visualization** → Displays rainfall distribution across states
- **Bar Chart** → Shows sum of rainfall by state and month
- **Line Chart** → Represents monthly rainfall trends
- **Pie Chart** → Displays rainfall distribution across states
- **Area Chart** → Shows rainfall and average rainfall trends
- **Table** → Displays detailed rainfall records

(These visuals are visible in your dashboard layout on page 1 and 2)

4.4 Filters and Interactivity

To make the dashboard interactive, slicers are added:

- State
- District
- Station Name
- Season

These filters allow users to dynamically explore data and perform customized analysis.

CHAPTER 4

4.5 Dashboard Design

The final dashboard is designed to present all insights in a structured layout:

- Left side → Filters (State, District, Station, Season)
- Center → Map visualization
- Right side → Trend charts
- Bottom → Detailed table and summary charts

This layout improves usability and makes the dashboard easy to understand.

4.6 Outcome

The methodology results in a fully interactive dashboard that helps users:

4. Analyze rainfall patterns
5. Compare states
6. Identify high rainfall regions
7. Understand monthly and seasonal trends

CHAPTER 5

IMPLEMENTATION

The implementation phase of the **“High Rainfall Dashboard”** project involves converting raw rainfall data into meaningful insights using Power BI. This includes data loading, transformation, creation of measures, and designing interactive visualizations.

5.1 Dataset Description

The dataset used in this project consists of rainfall-related information collected from weather records. As seen in the table (page 2 of your file), it includes the following fields:

- **State** – Name of the state
- **District** – District name
- **Station Name** – Weather station location
- **Date / Month / Year** – Time-related data
- **Season** – Seasonal classification (Winter, Monsoon, etc.)
- **Rainfall (mm)** – Amount of rainfall recorded
- **Average Temperature (avg_temp)** – Temperature value

This dataset is imported into Power BI using Excel/CSV format.

CHAPTER 5

5.2 Data Loading

- The dataset is loaded into Power BI using the **“Get Data”** option.
- Data is previewed and verified before processing.

5.3 Data Transformation (Power Query)

Using Power Query Editor, the dataset is cleaned and prepared:

- Removed null and duplicate values
- Renamed columns for clarity
- Converted data types (text → date, numeric)
- Extracted fields like **Month, Year, and Season**
- Ensured consistency across state and station names

5.4 Data Modeling

Since the project is simple, a **single-table model** is used:

- No complex relationships or ER diagram
- All data fields are directly used for visualization

This approach keeps the dashboard simple and efficient.

CHAPTER 5

5.5 DAX Measures (Key Calculations)

To generate insights, the following measures are created:

Total Rainfall = SUM(Rainfall[Rainfall])

Average Rainfall = AVERAGE(Rainfall[Rainfall])

Maximum Rainfall = MAX(Rainfall[Rainfall])

Minimum Rainfall = MIN(Rainfall[Rainfall])

These measures are used in charts and KPI visuals.

5.6 Dashboard Visualizations

The dashboard includes multiple visuals (as shown in your report):

Main Visuals:

- **Map Visualization**

  Displays rainfall distribution across different states using bubbles.

- **Bar Chart (Sum of Rainfall by State and Month)**

  Helps compare rainfall across states and months.

- **Line Chart (Sum of Rainfall by Month)**

  Shows rainfall trends over time.

CHAPTER 5

- **Pie Chart (Rainfall by State)**

  Represents percentage contribution of each state.

- **Area Chart (Rainfall and Average Rainfall)**

  Displays variation and comparison of rainfall values.

- **Table (Rainfall Record)**

  Provides detailed data view including station, state, and rainfall.

5.7 Filters / Slicers

To enhance interactivity, the following filters are added:

- State
- District
- Station Name
- Season

Users can dynamically filter data and analyze specific regions or time periods.

5.8 Dashboard Layout

The dashboard is designed in a structured way:

- Left side → Filters
- Center → Map visualization
- Right side → Trend charts
- Bottom → Detailed table and summary

This layout ensures better readability and user experience.

CHAPTER 5

5.9 Integration

- Data is imported from Excel/CSV
- All visuals are connected to the same dataset
- Filters dynamically update all visuals

5.10 Final Output

The implementation results in a fully interactive Power BI dashboard that:

- Displays rainfall patterns across India
- Allows state-wise and monthly analysis
- Helps identify high rainfall regions
- Provides clear and visual insights

CHAPTER 6

RESULT

The developed **“High Rainfall Dashboard”** successfully provides an interactive and visually rich analysis of rainfall patterns across different states. The dashboard transforms raw rainfall data into meaningful insights using various visualizations such as maps, bar charts, line graphs, and pie charts.

The **map visualization** clearly highlights the geographical distribution of rainfall, allowing users to easily identify regions with high and low rainfall. The **bar charts** enable comparison of rainfall across different states and months, while the **line chart** effectively shows rainfall trends over time. The **pie chart** represents the proportional contribution of each state to total rainfall, making it easier to understand distribution patterns.

Interactive **filters and slicers** such as state, district, station name, and season allow users to customize the analysis based on their requirements. This enhances user experience and enables detailed exploration of the dataset.

The dashboard also includes calculated measures like total rainfall and average rainfall, which provide quick insights into overall rainfall statistics. The integration of multiple visuals ensures that users can analyze both spatial and temporal patterns efficiently.

Overall, the project successfully achieves its objective of providing a simple, interactive, and informative platform for rainfall analysis. It helps users identify trends, compare regions, and gain valuable insights into rainfall behavior, which can be useful for environmental studies, agriculture planning, and water resource management.

CHAPTER 6

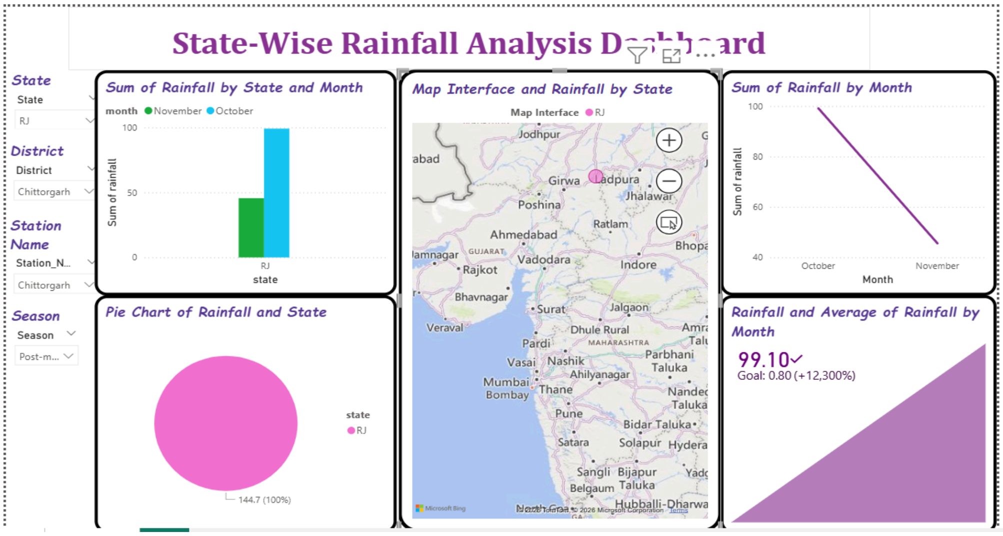


Fig 1 : State-Wise Rainfall Analysis Dashboard State-Rajasthan

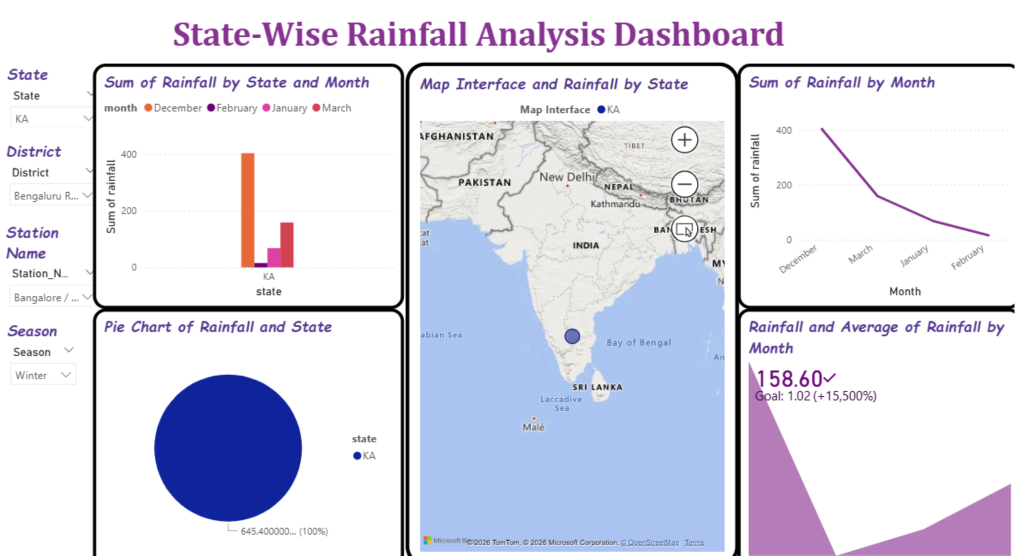
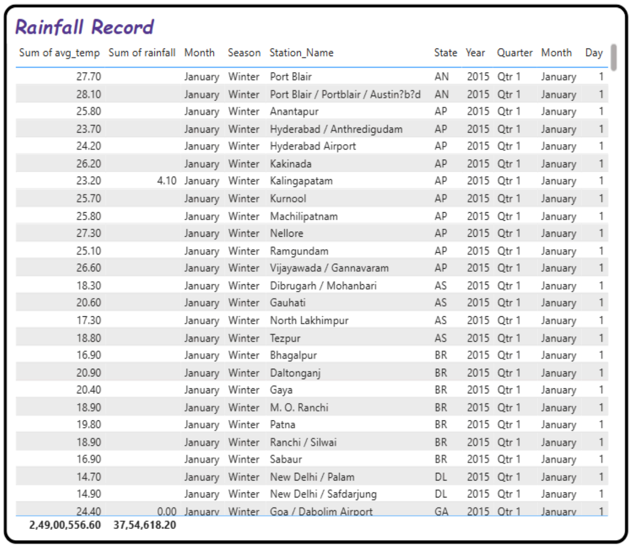


Fig 2 : State-Wise Rainfall Analysis Dashboard State-Karnataka

CHAPTER 6



Rainfall Record

Sum of avg_temp	Sum of rainfall	Month	Season	Station_Name	State	Year	Quarter	Month	Day
27.70		January	Winter	Port Blair	AN	2015	Qtr 1	January	1
28.10		January	Winter	Port Blair / Portblair / Austin?b?d	AN	2015	Qtr 1	January	1
25.80		January	Winter	Anantapur	AP	2015	Qtr 1	January	1
23.70		January	Winter	Hyderabad / Anthredigudam	AP	2015	Qtr 1	January	1
24.20		January	Winter	Hyderabad Airport	AP	2015	Qtr 1	January	1
26.20		January	Winter	Kakinada	AP	2015	Qtr 1	January	1
23.20	4.10	January	Winter	Kalingapatam	AP	2015	Qtr 1	January	1
25.70		January	Winter	Kurnool	AP	2015	Qtr 1	January	1
25.80		January	Winter	Machilipatnam	AP	2015	Qtr 1	January	1
27.30		January	Winter	Nellore	AP	2015	Qtr 1	January	1
25.10		January	Winter	Ramgundam	AP	2015	Qtr 1	January	1
26.60		January	Winter	Vijayawada / Gannavaram	AP	2015	Qtr 1	January	1
18.30		January	Winter	Dibrugarh / Mohanbari	AS	2015	Qtr 1	January	1
20.60		January	Winter	Gauhati	AS	2015	Qtr 1	January	1
17.30		January	Winter	North Lakhimpur	AS	2015	Qtr 1	January	1
18.80		January	Winter	Tezpur	AS	2015	Qtr 1	January	1
16.90		January	Winter	Bhagalpur	BR	2015	Qtr 1	January	1
20.90		January	Winter	Daltonganj	BR	2015	Qtr 1	January	1
20.40		January	Winter	Gaya	BR	2015	Qtr 1	January	1
18.90		January	Winter	M. O. Ranchi	BR	2015	Qtr 1	January	1
19.80		January	Winter	Patna	BR	2015	Qtr 1	January	1
18.90		January	Winter	Ranchi / Silwai	BR	2015	Qtr 1	January	1
16.90		January	Winter	Sabaur	BR	2015	Qtr 1	January	1
14.70		January	Winter	New Delhi / Palam	DL	2015	Qtr 1	January	1
14.90		January	Winter	New Delhi / Safdarjung	DL	2015	Qtr 1	January	1
24.40	0.00	January	Winter	Goa / Dabolim Airport	GA	2015	Qtr 1	January	1
2,49,00,556.60	37,54,618.20								

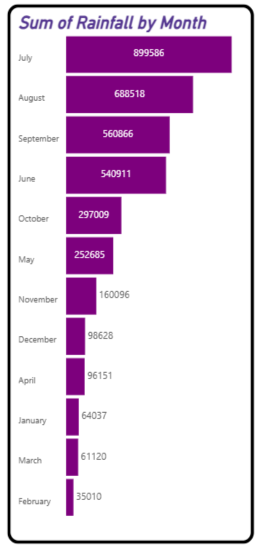


Fig 3 : Rainfall Dashboard Record and Monthly Analysis

CONCLUSION

The developed **“High Rainfall Dashboard”** successfully provides an interactive and visually rich analysis of rainfall patterns across different states. The dashboard transforms raw rainfall data into meaningful insights using various visualizations such as maps, bar charts, line graphs, and pie charts.

The **map visualization** clearly highlights the geographical distribution of rainfall, allowing users to easily identify regions with high and low rainfall. The **bar charts** enable comparison of rainfall across different states and months, while the **line chart** effectively shows rainfall trends over time. The **pie chart** represents the proportional contribution of each state to total rainfall, making it easier to understand distribution patterns.

Interactive **filters and slicers** such as state, district, station name, and season allow users to customize the analysis based on their requirements. This enhances user experience and enables detailed exploration of the dataset.

The dashboard also includes calculated measures like total rainfall and average rainfall, which provide quick insights into overall rainfall statistics. The integration of multiple visuals ensures that users can analyze both spatial and temporal patterns efficiently.

Overall, the project successfully achieves its objective of providing a simple, interactive, and informative platform for rainfall analysis. It helps users identify trends, compare regions, and gain valuable insights into rainfall behavior, which can be useful for environmental studies, agriculture planning, and water resource management.

REFERENCES

[1] Ministry of Earth Sciences, Government of India, *India Meteorological Department (IMD) Reports*, New Delhi, India, 2026.

[2] Government of India, *Open Government Data Platform India: Climate and Rainfall Data*, New Delhi, India, 2026.

[3] Microsoft Corporation, *Microsoft Power BI Documentation*, Redmond, WA, USA, 2026.

[4] Microsoft Corporation, *Data Analysis Expressions (DAX) Reference Guide*, Redmond, WA, USA, 2026.

[5] Microsoft Corporation, *Power Query (M Language) Documentation*, Redmond, WA, USA, 2026.

[6] World Meteorological Organization, *Global Climate Data and Monitoring Reports*, Geneva, Switzerland, 2025.

[7] Rainfall Dataset, *State-wise Rainfall Data for India*, Unpublished dataset, 2026.